

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Molecules Electroporated DNA: pLAR33 (an 18kB ds DNA plasmid), pAM401 (an 10.4 kB ds DNA plasmid)

Species Used *Enterococcus faecalis* JH2-2 and UV202

Before the Pulse

Cell Growth Medium Brain Heart Infusion (BHI) broth (Difco)

Growth Phase at Harvest O.D. (600) =1.0 (exponential)

Pre-pulse Incubation Buffer, cuvettes, and cells placed on ice

Wash Solution 1.5x EP (see notes)

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 25 °C

Electroporation Medium 1.5x EP

Cuvette Gap 0.2 cm

Voltage 2.5 kV

Cell Density Cells concentrated 140x

Volume of Cells 50 µl

Field Strength 12.5 kV/cm

DNA Concentration 1 ng DNA

DNA Resuspension Buffer TE (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 25 µF

Volume of DNA 2 µl

Resistor 200 Ω (Pulse Controller)

After the Pulse

Time Constant 5.6 to 5.8 msec

Outgrowth Medium BHI -Difco

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Wash solution: EP=1mM HEPES (pH7.4),1mM MgCl2, 0.5M sucrose

Outgrowth Temperature 37 °C

Length of Incubation 3 hours

Selection Method or Assay Used BHI agar plates with 10 µg / ml erythromycin

Electroporation Efficiency 2 x 10 (3) transformants / ng DNA

Per Cent Survival Not determined

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